

Asymmetric Impact of AI Matching on Influencer Marketing: Implications for Platform Revenue

Jessie Liu ¹ Yi Liu ²

¹ Johns Hopkins University

² University of Wisconsin-Madison

Social media platforms' big source of revenue



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- ▶ **34%** of Instagram users, **25%** of Facebook users, and **29%** of Twitter users claim they bought something because an influencer recommended it.
- ▶ **Influencer marketing** has become the largest revenue source (**47.0%**) for Kuaishou in 2020.

Most brands today have adopted influencer marketing

Share of US Marketers Using Social Media and Influencer Marketing, 2019-2022

% of total marketers

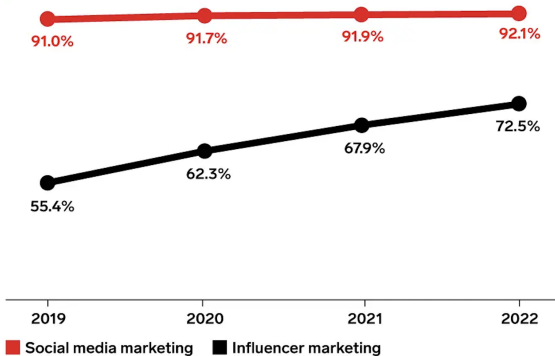


Figure: Source: eMarketer, InsiderIntelligence.com

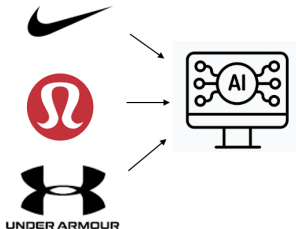
Challenge: Matching brands and influencers

“According to marketers, one of the most **frustrating challenges** associated with influencer marketing is choosing the **right influencers** with whom to work.” (Avery and Israeli, 2020)

- ▶ More than 500,000 active influencers on Instagram

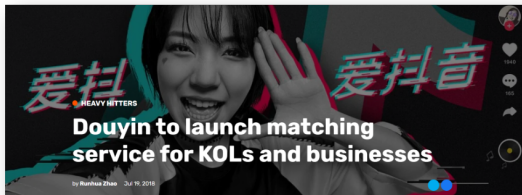
One solution: Brand-Influencer Matching AI

► Brand → Target Customers → AI recommended Influencers



@Username	Country/Region	Topics
 kingjames	 United States	Sports Basketball
 Karim Benzema karimbenzema	 United States	Sports Soccer
 House of Highlights houseofhighlights	 United States	Sports News Video Gaming Basketball Cooking Celebrities Media/News Company Publishers
 dwyanewade dwyanewade	 United States	Basketball Sports Athlete Creators & Celebrities
 John Cena johncena	 United States	Lifestyle Sports Geek
 Odell Beckham Jr obj	 United States	Sports American football Athlete
 Chris Bumstead cbum	 United States	Sports Training Coaching
 Lonzo Ball zo	 United States	Basketball Sports Personal Blog Creators & Celebrities

One solution: Brand-Influencer Matching AI



BrandConnect

This Paper

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 - ▶ Influencers can charge **higher prices**, and
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- ▶ AI does **lift** the **matching-efficiency** and thus the **sales** performance of influencer marketing, so
 - ▶ Influencers can charge **higher prices**, and
 - ▶ **More** influencers and marketers **participate**
- ▶ But...
 - ▶ AI can **change** the **competitive landscape** of influencers

Related Literature

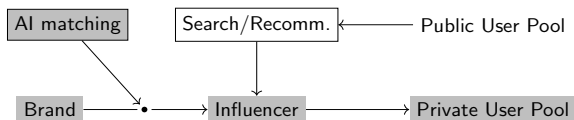
summary

- ▶ Influencer marketing
 - ▶ e.g. Pei and Mayzlin, 2019; Nistor and Selove, 2022

Related Literature

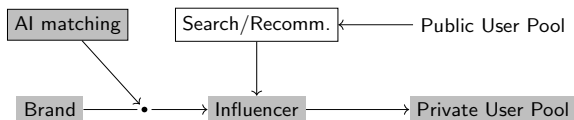
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- ▶ Influencer marketing
 - ▶ e.g. Pei and Mayzlin, 2019; Nistor and Selove, 2022
- ▶ Search frictions in online markets
 - ▶ e.g. Latzer et al., 2016; Fainmesser and Galeotti, 2021
 - ▶ **Our paper:**



Related Literature summary

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 - ▶ **Our paper:**



- ▶ Targeted advertising
 - ▶ e.g. Shaffer and Zhang, 1995; Brahim et al., 2011,
 - ▶ **Our paper:** “firms” (marketers) in our paper cannot directly target “consumers” (followers), but *indirectly* via influencers who set competitive prices.

Model

- ▶ **Consumers** with **heterogenous interests** in a representative product category are located on a **unit circle**
- ▶ **Influencers**
 - ▶ Broadness of follower interests: $n \sim U[0, \frac{1}{2}]$
 - ▶ Density of followers: $\rho > 0$

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 - ▶ Broadness of follower interests: $n \sim U[0, \frac{1}{2}]$
 - ▶ Density of followers: $\rho > 0$
- ▶ **Marketers**
 - ▶ Target consumers located at 0
 - ▶ Incur a fixed cost of IM campaign: $c \sim U[0, \frac{1}{2}]$
 - ▶ Obtain revenue M from working with **influencer** n with followers **centered around** x

$$M(x, n) = \int_{x-\frac{n}{2}}^{x+\frac{n}{2}} (\max \{0, \alpha - |\tilde{x} - 0|_d\}) \rho d\tilde{x},$$

where $\alpha > 0$ is the expected revenue from **target customer**,
 $|\tilde{x} - 0|_d \equiv \min \{1 - |\tilde{x}|, |\tilde{x}|\}$.

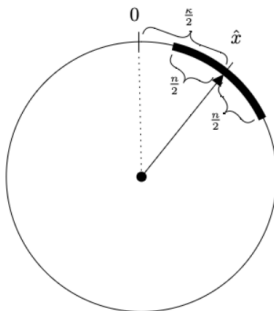
Model

► The platform

- offers an **AI-based service** that recommends to each marketer a set of influencers $n \in (0, \frac{1}{2}]$ with followers centered around

$$\hat{x} = 0 + \frac{\kappa}{2},$$

where κ is defined as the **bias** (and accuracy as $\lambda = 1 - |\kappa|$).



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3. Each **brand** chooses whether to participate and **buy** paid endorsement from **at most** one influencer
4. **Platform and influencers share** the price of influencer marketing campaigns at a **commission rate** of $\delta \in (0, 1)$ such that the platform obtains a revenue of

$$\pi = \int_{n \in \mathcal{N}} \delta p(n) dn,$$

where $\mathcal{N}$ denotes the set of influencers who endorse a brand.

Competitive Pricing of Influencers

综合排序 粉丝数排序 报价排序 粉丝属性排序



沈阳山胖 (善良保安) id:shenyangsanpang666
粉丝数: 345万
网红作品分类: 搞笑

Paid sponsorship price:
337,500 RMB (3 days)

视频推广报价: 27000元 (3天)

1天: 15660元 7天: 33750元

加入选择列表



大胃王阿浩 727 id:292451142
粉丝数: 1650万 # followers: 16.50 million

网红作品分类: 搞笑, 美食 Influencer category: comedy, gourmet

视频推广报价: 337500元 (3天)

1天: 195750元 7天: 421880元

加入选择列表



铁娃 二弟 黄金搭档 id:239167604
粉丝数: 453万

网红作品分类: 搞笑

视频推广报价: 67500元 (3天)

1天: 39150元 7天: 84380元

加入选择列表



韩兆导演 (条子哥) id:tiaoziye
粉丝数: 1701万
网红作品分类: 搞笑

Paid sponsorship price:
87,750 RMB (3 days)

视频推广报价: 108000元 (3天)

1天: 62640元 7天: 135000元

加入选择列表



岳老板 (梵·高) id:fangao888
粉丝数: 842万 # followers: 8.42 million

网红作品分类: 搞笑 Influencer category: comedy

视频推广报价: 87750元 (3天)

1天: 50900元 7天: 109690元

加入选择列表

Equilibrium Concept

Definition 1

At any given λ , an equilibrium is a strategy profile such that

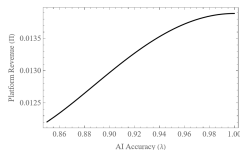
- i $p^*(n)$ maximizes **influencer** n 's expected utility;
- ii every **participating marketer** $c \in \mathcal{C}$ that chooses influencer $n \in \mathcal{N}$ prefers buying an endorsement from her at $p^*(n)$ to buying from other influencers at their equilibrium prices;
- iii every **non-participating** marketer $c \notin \mathcal{C}$ obtains non-positive utility from buying an endorsement from any influencers at their respective prices.
- iv Market clears such that:

$$\int_{n \in \mathcal{N}} dn = \int_{c \in \mathcal{C}} dc. \quad (1)$$

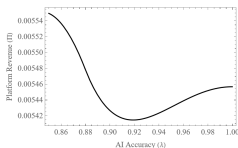
Main Result

Proposition 1

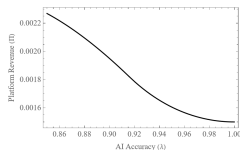
The platform's revenue (Π) does not always increase in the AI prediction accuracy (λ). In particular, we have $\frac{\partial \Pi}{\partial \lambda} \leq 0$ when the followers' density (ρ) falls below a threshold.



(a) $\rho = 30$



(b) $\rho = 21$



(c) $\rho = 15$

Mechanism

- ▶ Heterogenous effect of AI improvement → competition among influencers

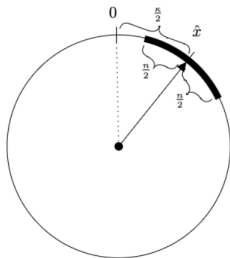


Figure: **Niche** influencers are **more** sensitive to AI accuracy



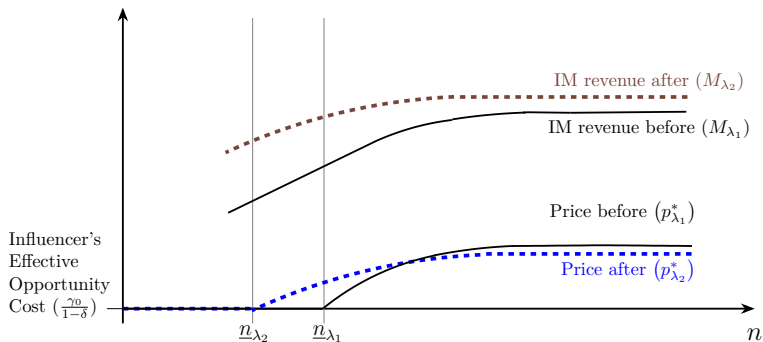
Mechanism

- ▶ Participating influencers: $n \geq \underline{n}$
- ▶ Participating brands: $c \leq \bar{c}$

$$\begin{aligned} \frac{\partial \Pi}{\partial \lambda} = & \delta \int_{\underline{n}}^{\frac{1}{2}} \left(\underbrace{\underbrace{\frac{\partial M}{\partial \lambda}}_{\text{matching enhancement (+)}} \underbrace{-\frac{\partial \bar{c}}{\partial \lambda}}_{\text{competition (-)}}}_{\frac{\partial p^*(n)}{\partial \lambda}: \text{effect on equilibrium prices}} \right) dn \\ & + \delta \underbrace{\left(-\frac{\partial \underline{n}}{\partial \lambda} \right) \cdot p^*(\underline{n})}_{\text{market expansion (+)}} \end{aligned}$$

Mechanism

- Suppose AI accuracy increases from λ_1 to λ_2 ...



Discussions on Competition

- ▶ Our paper focuses on “competition” among influencers
 - ▶ working for **marketers with the same target consumer segment**
 - ▶ working with **up to only one** marketer/brand
 - ▶ This assumption is reasonable when marketers compete with each other in the same product category
 - ▶ Our analysis remains valid as long as all influencers are endowed with *equal* probabilities to work with multiple brands

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 - ▶ Our analysis remains valid as long as all influencers are endowed with *equal* probabilities to work with multiple brands
- ▶ If general influencers are allowed to promote multiple brands with different target consumers due to their diverse follower base
 - ▶ **Our conjecture:** **enhanced** matching could lead to higher *within-product-* category competition among influencers → it spills over to competition in other product categories → **exacerbate** the potential decline in platform revenue.

Main Takeaway

- ▶ Improving a brand-influencer matching AI is not always in a platform's best interest
- ▶ Optimally adjusting commission rate can **mitigate** the negative effect of AI backfire, but may not eliminate it if **larger** influencers face **higher** opportunity costs

Contribution

“All models are wrong, but some are useful. ” – George Box

- ▶ What is unique about “influencer marketing” per se?
 - ▶ Matching technology **improves** the **fit** between influencers and marketers, but it does not change the **composition** of influencers' followers
 - ▶ **Micro-founding** how technology affects the brand-influencer matching process helps us uncover its **asymmetric** impact on different types of influencers.

literature